

Theoretical Physics for Mathematicians

Home Work 1

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September 16, 2026

Newtonian Space-Time and the Galilei Group

Exercise 1 ($\star$ The Galilei group). For $a \in \mathbb{R}$, $\vec{b}, \vec{v} \in \mathbb{R}^3$ and $R \in SO(3)$ define the map

$$g_{(a, \vec{b}, \vec{v}, R)} : \mathbb{R} \times \mathbb{R}^3 \longrightarrow \mathbb{R} \times \mathbb{R}^3, \quad (t, \vec{x}) \longmapsto (t + a, R\vec{x} + \vec{v}t + \vec{b}).$$

- (a) Show that the composition of two such maps is again of this form, and derive the composition law

$$g_{(a_2, \vec{b}_2, \vec{v}_2, R_2)} \circ g_{(a_1, \vec{b}_1, \vec{v}_1, R_1)} = g_{(a_1 + a_2, R_2 \vec{b}_1 + a_1 \vec{v}_2 + \vec{b}_2, R_2 \vec{v}_1 + \vec{v}_2, R_2 R_1)}.$$

Deduce that the set $\mathcal{G}$ of all these maps is a group, compute the inverse of $g_{(a, \vec{b}, \vec{v}, R)}$, and determine $\dim \mathcal{G}$.

- (b) Let

$$N = \left\{ g_{(a, \vec{b}, 0, I)} \mid a \in \mathbb{R}, \vec{b} \in \mathbb{R}^3 \right\}$$

be the subgroup of space-time translations. Show that N is normal in $\mathcal{G}$ and that $\mathcal{G}/N \cong SO(3) \ltimes \mathbb{R}^3$. Conclude that $\mathcal{G} \cong \mathbb{R}^4 \ltimes (SO(3) \ltimes \mathbb{R}^3)$.

- (c) Show that the boosts

$$B = \left\{ g_{(0, 0, \vec{v}, I)} \mid \vec{v} \in \mathbb{R}^3 \right\}$$

form an abelian subgroup isomorphic to $(\mathbb{R}^3, +)$, but that B is not normal in $\mathcal{G}$.

- (d) Let $\vec{x} : I \rightarrow \mathbb{E}^3$ be a trajectory and let $\tilde{\vec{x}}$ be its image under $g_{(a, \vec{b}, \vec{v}, R)}$. Show that

$$\ddot{\vec{x}} \equiv 0 \quad \text{if and only if} \quad \ddot{\tilde{\vec{x}}} \equiv 0.$$

Hence the notion of an inertial system is well defined on the whole group $\mathcal{G}$.

(a)

Show that the Galilean transformations are closed under composition.

Take

$$\begin{aligned} g_{(a_1, \hat{b}_1, \hat{v}_1, R_1)}(t, \bar{x}) &= (t + a_1, R_1 \bar{x} + \hat{v}_1 t + \hat{b}_1), \\ g_{(a_2, \hat{b}_2, \hat{v}_2, R_2)}(t, \bar{x}) &= (t + a_2, R_2 \bar{x} + \hat{v}_2 t + \hat{b}_2). \end{aligned}$$

Then

$$\begin{aligned} g_{(a_2, \hat{b}_2, \hat{v}_2, R_2)} \circ g_{(a_1, \hat{b}_1, \hat{v}_1, R_1)}(t, \bar{x}) &= g_{(a_2, \hat{b}_2, \hat{v}_2, R_2)}(t + a_1, R_1 \bar{x} + \hat{v}_1 t + \hat{b}_1) \\ &= (t + a_1 + a_2, R_2[R_1 \bar{x} + \hat{v}_1 t + \hat{b}_1] + \hat{v}_2(t + a_1) + \hat{b}_2) \\ &= (t + a_1 + a_2, (R_2 R_1) \bar{x} + (R_2 \hat{v}_1 + \hat{v}_2)t + R_2 \hat{b}_1 + \hat{v}_2 a_1 + \hat{b}_2) \\ &= (t + a_1 + a_2, (R_2 R_1) \bar{x} + (R_2 \hat{v}_1 + \hat{v}_2)t + (R_2 \hat{b}_1 + \hat{v}_2 a_1 + \hat{b}_2)) \\ &= g_{(a_1 + a_2, R_2 \hat{b}_1 + \hat{v}_2 a_1 + \hat{b}_2, R_2 \hat{v}_1 + \hat{v}_2, R_2 R_1)}. \end{aligned} \quad (*)$$

Now $R_1, R_2 \in SO(3) \Rightarrow R_2 R_1 \in SO(3)$. Also: $R_2 \bar{v}_1 + \bar{v}_2 \in \mathbb{R}^3$, as are $R_2 \bar{b}_1 + \bar{b}_2 \in \mathbb{R}^3$.

Hence (*) is also a Galilean transformation.

The identity element of this group will leave $(t, \bar{x})$ fixed. Thus:

$$\begin{aligned} g_{(a, \bar{b}, \bar{v}, R)}(t, \bar{x}) &= (t + a, R \bar{x} + \bar{v} t + \bar{b}) \\ &= (t, \bar{x}). \end{aligned}$$

It is clear that: $a = 0, \bar{v} = \bar{0}, \bar{b} = \bar{0}$ and $R = I$.

Hence $g_{(0, \bar{0}, \bar{0}, I)}$ is the identity element for the group $\mathcal{G}$.

Now let's find the inverse transformation for $g_{(a, \bar{b}, \bar{v}, R)} \in G$. Such that:

$$g_{(c, \bar{d}, \bar{u}, T)} \circ g_{(a, \bar{b}, \bar{v}, R)} = g_{(0, \bar{0}, \bar{0}, I)}. \quad (1)$$

$$g_{(a, \bar{b}, \bar{v}, R)} \circ g_{(c, \bar{d}, \bar{u}, T)} = g_{(0, \bar{0}, \bar{0}, I)}. \quad (2)$$

From (*) above we know that

$$\begin{aligned} g_{(c, \bar{d}, \bar{u}, T)} \circ g_{(a, \bar{b}, \bar{v}, R)} &= g_{(a+c, T\bar{b}+\bar{u}a+\bar{d}, T\bar{v}+\bar{u}, TR)} \\ &= g_{(0, \bar{0}, \bar{0}, I)}. \end{aligned}$$

Clearly:

$$\begin{aligned} a + c = 0 &\Rightarrow c = -a, \\ TR = I &\Rightarrow T = R^{-1} \quad (\text{possible as } R \in SO(3)), \\ T\bar{v} + \bar{u} = \bar{0} &\Rightarrow \bar{u} = -T\bar{v} = -R^{-1}\bar{v}, \\ T\bar{b} + \bar{u}a + \bar{d} = \bar{0} &\Rightarrow R^{-1}\bar{b} - (R^{-1}\bar{v})a + \bar{d} = \bar{0} \\ &\Rightarrow \bar{d} = R^{-1}(\bar{v}a - \bar{b}). \end{aligned}$$

The same result applies if we take (2). Hence

$$g_{(a,\bar{b},\bar{v},R)}^{-1} = g_{(-a, R^{-1}(\bar{v}a - \bar{b}), -R^{-1}\bar{v}, R^{-1})}.$$

The above three properties: closure, inverse and identity show that $\mathcal{G}$, the set of Galilean transformations, is indeed a group.

(b)

To demonstrate that: $N = \{g_{(a,\bar{b},\bar{0},I)} \mid a \in \mathbb{R}, \bar{b} \in \mathbb{R}^3\}$ is normal in $\mathcal{G}$ we need to prove that: given any $g \in \mathcal{G}$ and $n \in N$, that:

$$g \circ n \circ g^{-1} \in N$$

This is just subscript manipulation as we can use equation (*) repeatedly. Take $n = g_{(a,\bar{b},\bar{0},I)}$ and $g = g_{(a_2,\bar{b}_2,\bar{v}_2,R_2)}$. Then:

$$\begin{aligned} g_{(a_2,\bar{b}_2,\bar{v}_2,R_2)} \circ n \circ g_{(a_2,\bar{b}_2,\bar{v}_2,R_2)}^{-1} &= g_{(a_2,\bar{b}_2,\bar{v}_2,R_2)} \circ g_{(a,\bar{b},\bar{0},I)} \circ g_{(-a_2,R_2^{-1}(\bar{v}_2 a_2 - \bar{b}_2), -R_2^{-1}\bar{v}_2, R_2^{-1})} \\ &= g_{(a_2+a, R_2\bar{b} + a\bar{v}_2 + \bar{b}_2, R_2\bar{v} + \bar{v}_2, R_2I)} \circ g_{(-a_2, R_2^{-1}(\bar{v}_2 a_2 - \bar{b}_2), -R_2^{-1}\bar{v}_2, R_2^{-1})} \\ &= g_{(a_2+a, R_2\bar{b} + a\bar{v}_2 + \bar{b}_2, R_2\bar{v} + \bar{v}_2, R_2I)} \circ g_{(-a_2, R_2^{-1}(\bar{v}_2 a_2 - \bar{b}_2), -R_2^{-1}\bar{v}_2, R_2^{-1})} \\ &= g_{(a,0,z,I)} \end{aligned}$$